Cheemi Technology Co., Ltd

DATA SHEET
Hall Effect Current Sensor

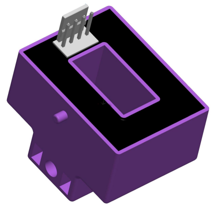
PN: CHK_BSR5S6L-02	I_{PN}=50-600A

Feature

- Open- loop
- Supply voltage: DC +5.0V
- Capable measurement of currents: DC, AC,pulse with galvanic isolation between primary circuit and secondary circuit.

Advantages

- Easy installation
- No insertion losses
- Low power consumption
- Wide current measuring range
- High immunity to external interference
- Can be customized



Applications

- Inverter applications
- AC/DC variable-speed drive
- Uninterruptible Power Supplies (UPS)
- Switched Mode Power Supplies (SMPS)
- Frequency drive control home appliances

CE RoHS

Electrical data: （Ta=25°C±5℃，Vc=+5.0VDC, R_L=2KΩ，C_L=10000PF）

Parmeter \ Ref	CHK50 BSR5S6L-02	CHK100 BSR5S6L-02	CHK200 BSR5S6L-02	CHK300 BSR5S6L-02	CHK400BS R5S6L-02	CHK600 BSR5S6L-02
Rated input Ipn(A)	50	100	200	300	400	600
Measuring range Ip(A)	0～±150	0～±300	0～±600	0～±900	0～±900	0～±900
Rated output Vo(V)	@Ip=±Ipn	±0.625±1%				
Zero voltage Vo(V)	@IP=0, T=25°C	2.500±0.5%				
Reference voltage (V)	2.5±0.5%					
Supply voltage Vc(V)	+5.0 ±5%					
Zero offset voltage Voe(mV)	@IP=0,T=25°C	< ±20				
Magnetic offset （mV）	±15	±10				
Offset drift (mV/℃)	≤±1.0	≤±0.5				
Output drift (mV/℃)	≤±1.0	≤±0.5				
Linearity error εr(%FS)	@Ip=0-±Ipn	< 1.0				
Response time tra(μs)	@50A/μS, 10%-90%	< 3.0				
Power consumption Ic(mA)	15					
Insulation voltage Vd(Vd)	@50/60Hz, 1min,AC	2.5				

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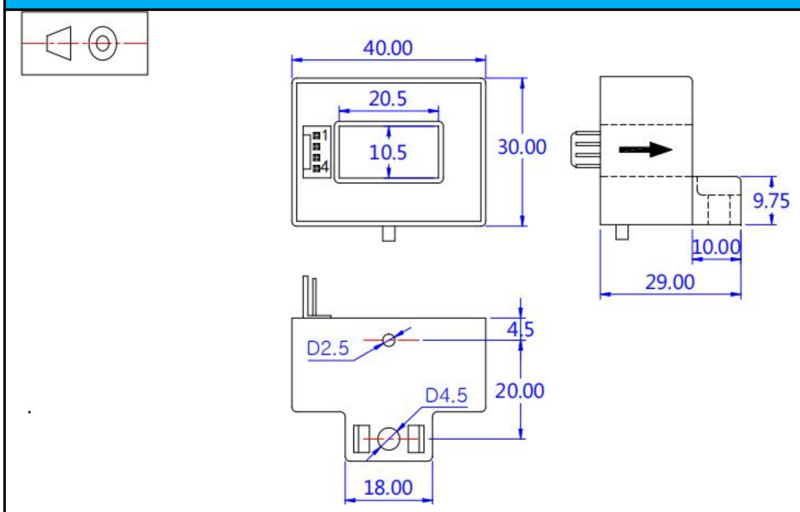
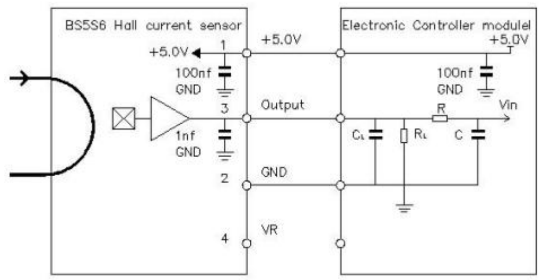
General data:

Parameter	Value
Operating temperature Ta(℃)	-40 ~ +85
Storage temperature Ts(℃)	-40~ +125
Mass M(g)	65
Plastic material	PBT G30/G15，UL94- V0;
Standards	IEC60950-1:2001
	EN50178:1998
	SJ20790-2000

Dimensions(mm):



Connection



General tolerance

General tolerance:< ±0.5mm

Primary through-hole： 10.5*20.5±0.3

Connection of secondary : 2510-4P

Plastic case :PBT GF30

Magnetic core: Iron silicon alloy

Electrical terminal:Brass tin platted

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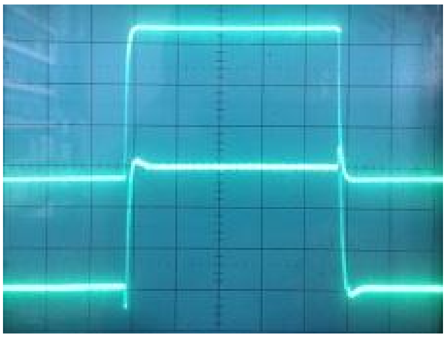
Characteristics chart:

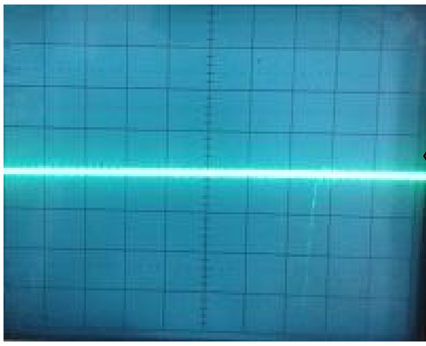
Pulse current signal response characteristic



⇦ Input signal

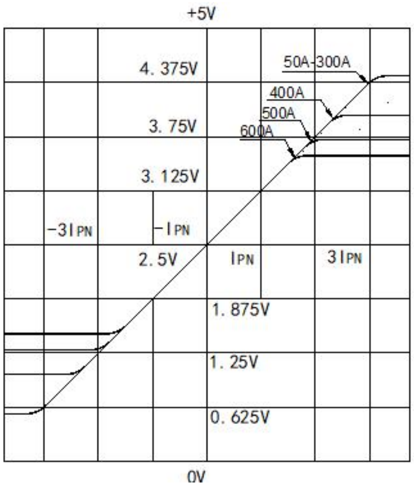
⇦ Output signal

Effects of impulse noise



⇦ Output voltage

Input current-Output Voltage characteristic



Remarks:

- When the current goes through the primary pin of a sensor, the voltage will be measured at the output end.
- Custom design is available for the different rated input current and the output voltage.
- The dynamic performance is the best when the primary hole if fully filled with.
- The primary conductor should be <100°C.

WARNING : Incorrect wiring may cause damage to the sensor.

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